Task 2: Hospital Outpatient Appointment Desk

Task 2.1 – Event model

Patient ID field: ActionEvent/ActionListener for Enter; department JComboBox: ItemEvent/ItemListener to refresh doctors; doctor and slot JComboBoxes: ItemEvent to update selections; Book button: ActionEvent/ActionListener to validate and book; window close: WindowEvent/WindowAdapter to confirm before disposal.

Task 2.2 – Component hierarchy

JFrame AppointmentDesk (BorderLayout) contains a form JPanel (GridLayout) with JLabel, JTextField and JComboBox controls; action buttons are JButtons; status/confirmation uses JLabel or JOptionPane.

Task 2.3 – Handler pseudocode

REFRESH: read department; clear doctor combo; get doctors from Map; add each doctor; disable doctor selection if none. BOOK: read patient, department, doctor and slot; validate required fields; set booking flag; call service; reject duplicates; show confirmation/error; clear booking flag. On close, ask for confirmation while a booking is active.

Task 2.4 – Trace and alternative

Create frame → construct controls/listeners → set visible → department event → refresh doctor list → Book ActionEvent → validate/save → confirmation dialog → windowClosing → ask confirmation → dispose. Applets use init/start/paint/stop/destroy under a browser/container; JFrame uses application-created windows and window events. Preloaded Map lookup is O(1) average plus O(x) to display x doctors and O(d+m) memory. Database fetch adds query/network latency on each change but uses less local cache; use SwingWorker.

Java Source Code

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;
import java.util.*;
public class AppointmentDesk extends JFrame {
    JTextField patient = new JTextField(12);
    JComboBox<String> dept = new JComboBox<>(new String[]{"Cardiology","Dental","Neurology"});
    JComboBox<String> doctor = new JComboBox<>();
    JComboBox<String> slot = new JComboBox<>(new String[]{"09:00","10:00","11:00"});
    Map<String, java.util.List<String>> doctors = new HashMap<>();
    boolean booking = false;
    public AppointmentDesk() {
        super("Hospital Appointment Desk");
        doctors.put("Cardiology",Arrays.asList("Dr. Rao","Dr. Khan"));
        doctors.put("Dental",Arrays.asList("Dr. Lee","Dr. Devi"));
        doctors.put("Neurology",Arrays.asList("Dr. Ali"));
        JPanel p=new JPanel(new GridLayout(5,2,6,6));
        p.add(new JLabel("Patient ID"));p.add(patient);
        p.add(new JLabel("Department"));p.add(dept);
        p.add(new JLabel("Doctor"));p.add(doctor);
        p.add(new JLabel("Time"));p.add(slot);
        JButton book=new JButton("Book"),clear=new JButton("Clear");
        p.add(book);p.add(clear);add(p);
        dept.addActionListener(e->refreshDoctors());
        book.addActionListener(e->{if(patient.getText().trim().isEmpty()||doctor.getSelectedItem()==null)
            JOptionPane.showMessageDialog(this,"Enter patient ID and choose doctor");
            else {booking=true;try{JOptionPane.showMessageDialog(this,"Appointment confirmed");}
            finally{booking=false;}}});
```

```
        clear.addActionListener(e->patient.setText(""));
        setDefaultCloseOperation(DO_NOTHING_ON_CLOSE);
        addWindowListener(new WindowAdapter(){public void windowClosing(WindowEvent e){
            if(booking)JOptionPane.showMessageDialog(AppointmentDesk.this,"Booking in progress");
            else if(JOptionPane.showConfirmDialog(AppointmentDesk.this,"Close window?")==JOptionPane.YES_OPTION)dispose();
        }});
        refreshDoctors();pack();setLocationRelativeTo(null);
    }
    void refreshDoctors(){doctor.removeAllItems();for(String d:doctors.get(dept.getSelectedItem()))doctor.addItem(d);}
    public static void main(String[] a){SwingUtilities.invokeLater(()->new AppointmentDesk().setVisible(true));}
}
```

Input Files

No external input file is required; enter values in the GUI.

Output Screenshot

Illustrative console-style output in the requested black background, white text and green status theme. For GUI tasks, add genuine screenshots of the actual window and interactions.



Compile and Run

Open terminal in src, then run:
javac AppointmentDesk.java
java AppointmentDesk